DGMS 2019 • SECTION M: REMOTE SENSING & RADIOMETRY

Question 91: Radiometric Quantities in Remote Sensing: Irradiance vs. Exitance

Official Examination Guide • Standard SI Radiometric Terminology, Radiant Flux & Incident Energy Density

OFFICIAL EXAMINATION QUESTION

The amount of radiant energy at wavelengths (λ) incident upon a surface per unit area of that surface is known as

(a) Radiant energy (Qλ)	(b) Radiant flux (Φλ)
(c) Irradiance (Eλ) CORRECT ANSWER	(d) Radiant exitance (Mλ)

1. Executive Summary & Core Principle

In optical physics and remote sensing (Lillesand, Kiefer & Chipman, *Remote Sensing and Image Interpretation*, Chapter 1; Jensen, *Introductory Digital Image Processing*), radiometric terms describe how electromagnetic radiation is emitted, transferred, and received:

- **Irradiance (E_λ or E):** The measure of radiant flux (or radiant energy per unit time) **incident upon a surface per unit area** of that surface:

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  Eλ = dΦλ / dA
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  It is expressed in Watts per square meter ($\text{W}\cdot\text{m}^{-2}$ or spectrally as $\text{W}\cdot\text{m}^{-2}\cdot\mu\text{m}^{-1}$). In Earth

observation, solar radiation arriving at the Earth's surface or at the top of the atmosphere is referred to as solar irradiance (E).

- **Radiant Energy (Q_λ):** The total energy carried by electromagnetic waves, measured in Joules (J).
- **Radiant Flux (Φ_λ):** The time rate of flow of radiant energy ($\Phi = dQ/dt$), measured in Watts ($\text{W} = \text{J/s}$).
- **Radiant Exitance (M_λ, formerly Radiant Emittance):** The rate of radiant energy (radiant flux) **emitted or leaving from a surface per unit area** ($M = d\Phi / dA$, also in $\text{W}\cdot\text{m}^{-2}$).
- **Radiance (L_λ):** The radiant flux per unit projected source area per unit solid angle ($\text{W}\cdot\text{m}^{-2}\cdot\text{sr}^{-1}$). Radiance is the fundamental quantity measured by airborne and satellite sensor detectors.

Because the question specifies the radiant energy per second (radiant flux) *incident upon a surface per unit area*, the correct term is **(c) Irradiance (E_λ)**.

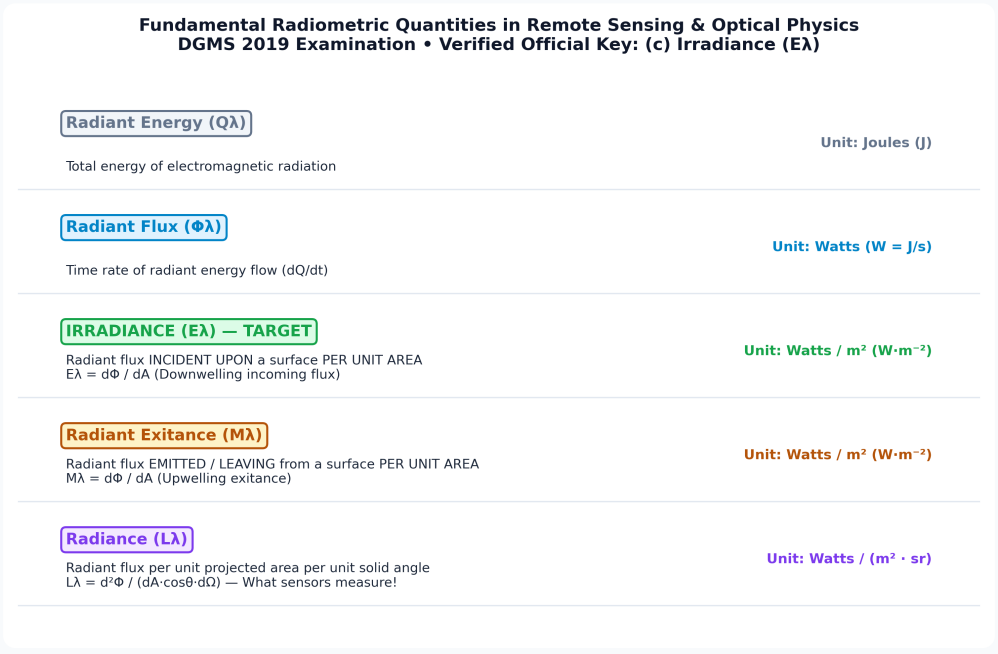


Figure 91.1: Hierarchy and definitions of fundamental radiometric quantities in optical remote sensing, contrasting downwelling Irradiance (E_λ) with upwelling Exitance (M_λ) and Radiance (L_λ) (Grounded in Lillesand & Kiefer).

2. Fundamental SI Radiometric Quantities Table

Radiometric Quantity	Symbol	Mathematical Definition	SI Unit	Physical Meaning in Remote Sensing
Radiant Energy	Q or Q_λ	Total electromagnetic energy	Joule (J)	Energy accumulated over time.
Radiant Flux	Φ or $ \Phi_\lambda$	$\Phi = dQ / dt$	Watt ($\text{W} = \text{J/s}$)	Time rate of radiant energy flow.
Irradiance	E or E_λ	**$E = d\Phi / dA$ (Incident)**	**$\text{W} \cdot\text{m} ^{-2}$**	**Radiation arriving at/ incident upon a target surface.**
Radiant Exitance	M or M_λ	$M = d\Phi / dA$ (Emitted)	$\text{W} \cdot\text{m} ^{-2}$	Radiation emitted or reflected away from a target surface.
Radiance	L or L_λ	$L = \frac{d^2\Phi} {dA \cos\theta d\Omega}$	$\text{W} \cdot\text{m} ^{-2}\cdot\text{sr} ^{-1}$	Directional brightness measured by remote sensing sensor telescopes.

3. Evaluation of Options

- **(a) Radiant energy (Q_λ): INCORRECT.** Represents total energy in Joules, not energy per unit area.
- **(b) Radiant flux (Φ_λ): INCORRECT.** Represents total energy rate in Watts (J/s), not normalized by surface area.
- **(c) Irradiance (E_λ): CORRECT ANSWER.** By definition, irradiance is radiant flux incident upon a surface per unit area (W/m^2).
- **(d) Radiant exitance (M_λ): INCORRECT.** Represents radiant flux *leaving or emitted* from a surface, not incident upon it.

> **Key Takeaway:** Energy incident upon a surface per unit area = **Irradiance (E_λ)**; Energy leaving a surface per unit area = **Radiant Exitance (M_λ)**. Hence, **(c)** is the correct choice.

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